Computers Today

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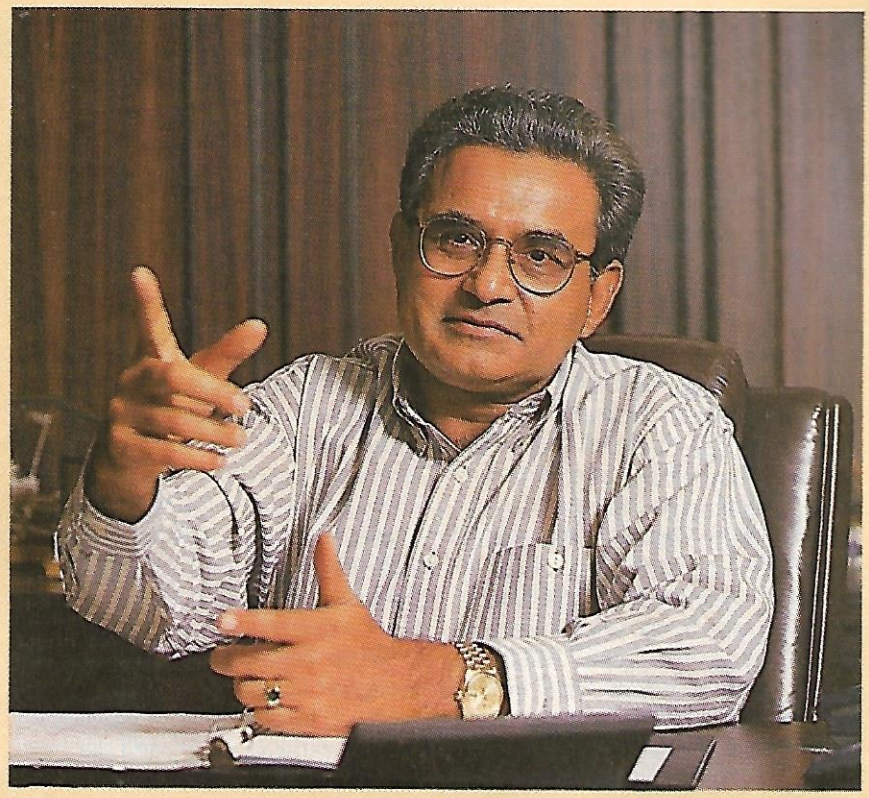
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THE TENTH YEAR

Pivotal to the Indian operations of the Tandon band of companies, is the golf-loving post graduate electrical engineer from Purdue University, chairman, M.L. Tandon, 55. His expertise, coupled with his brothers' acumen and direct clout in the West, is the backbone of the group operations. But, the ex-IBMer is generally not vociferous. He spoke to *Computers Today*.



"Commitment is What Carries You Through."

● What was your first breakthrough?

● Computer business was the domain of very large R&D-oriented companies when we started, and the revolution for PCs was coming. We were a participant of that, since Tandon, USA had developed a recording head which was very competitive, and with our ability of bringing that to India, and with Indian techniques and infrastructure we were able to make that still cheaper. So we had two advantages—a very good design and a very good manufacturing base—and that is how we started. It was breakthrough for the computer industry. During the process, the industry started realising that the market is very price elastic and if you bring the price down, the market increases.

Earlier, high prices of computers didn't matter much because those who needed it, had to buy it. But when prices started coming down, more started saying: "Yes, I can afford it." Hardware prices came down and software companies came into the picture.We started in 1978 and right through we have been viewing India as a low-cost manufacturing base.

● How were you different from the others, then?

● We did not follow the herd behaviour. Companies in India were making money by imports. You, either had a license or became a company's representative. Some even did the screwdriver technology business of importing and manufacturing and then, selling it. We said that India should be a net exporter and that's when we selected the product and the site as SEEPZ, paving the entry of foreign exchange and technology. And I think we have been very successful in that.

● Why is there practically no competition for you, in the country today?

● Here, the return on investment is not better in exports, in the sense that the amount of effort you put in is not proportionate to the returns you get. The number of hours put in and the hard work our team does—if we had put in the local market—we would have been a lot more successful. Then, you must have the capability and reputation to participate in the international market. No one is going to buy from you just because you are there. You have to have the ability to convince the customer that I am serious and come what may, I will provide prompt and better services. I have personally taken a shipment in my car to the airport to see the customer does not get hurt, because there was a transporters' strike. Some people may say, "...how does it matter when I can send

it tommorrow." So a high degree of customer satisfaction has to be part of the system and it needs total committment from all sections of the staff.

● A brief on your company's history...

● The first product we started manufacturing was cable harness assemblies. We saw in India, the potential to be a net exporter. And perhaps a part of that was my personal committment. I wanted to do something in my country and say, "Yes, India can do it." There is more of that than anything else. Opportunities may come and go, but your personal committment is what carries you through.

Next came harness assemblies which was supposed to be a very mundane job. Then we went on to making heads in which we were very successful. We made 200,000 heads per month, bulk of which went to IBM. After that we started manufacturing stepper motors. We were making heads, the most important component of the floppy drive. The second most expensive and important component was the motor. We took the technology from a Hitachi subsidiary and formed a company by the name of Tandon Motors. This technology changed and the business went down. Now, we are in the process of coming out with a new series of motors.

● What after heads?

● We got into making floppy disk drives in very large volumes. Tandon Corp., USA was one of the companies which pioneered the floppy disk drive at a low cost. People used to laugh when we said floppies would be available at 30 dollars when they cost $400 to 500 to start with.

When I made heads I supplied to just about everybody in the business. Today I have the best of companies as customers for the head business. Nobody in India can claim to have such a client list, and even for the US or other developed countries it is a very impressive list. To create in that kind of an environment where even Indians did not believe in India, what to say of foreigners, is what I call achievement. And it is matter of pride to an Indian.

● What were the reasons for your revenues dipping in 1991-92?

● Everything has a limited life and goes through changes. Success lies in understanding them and in making the changes. But, it all depends on market conditions. 1991-92 was basically a transition period. The product changed, and in business it takes time to get the quality certificate from the customer. Introducing new products takes time. It happens that we do not react in time and technology changed. So, we had to work hard enough to spring back. And we are also subject to international business conditions. If the market goes down, we obviously go down with it and in addition, it also depends on our performance. If the customers are not happy they find somebody else.

What affected most, was India's going through a crisis three years ago. We were on the brink of bankruptcy when it came to Forex, and our business suffered heavily. Since, we do not have the necessary input (raw material) in the country, we have to depend on it from outside. We could not import anything. So, we could not export anything either, and the revenues fell while reputation went down. I always worry more about reputation than revenues, because revenues can be built up but reputation takes a long time to build.

● Which are you focus areas now?

● We used to make floppies. We made computers but we are making less these days. Now we are focussing on power supplies, heads and motors.

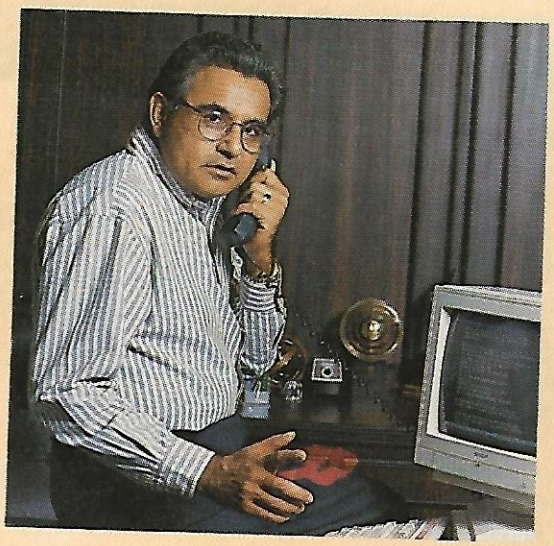
● You make components of the CPU for the computer and assemble the whole system. Why are you not getting into manufacturing monitors and keyboards?

● I am not a 'me too' kind of a guy who says, "Yes, I also make the monitor." If I do it, I would want to do it right and on a large-scale. And right now we do not have the space available. Recently, we have set up a unit in NEPZ. I am seriously contemplating a new product in the Madras factory and if this comes through, we would probably double our revenues in one year.

● Have your years at IBM helped you in your business?

● It is not so much of technology that I got from IBM. The sense of committment and professionalism to customers was more of what I got from it: how to manage, how to handle and, not to forget, the exposure. I cannot bank on a technology that changes. If I did, I would be out of this business. But I did get the skills one gets from working in a multinational environment. Respect for an individual was one of the big things I learnt there. Pay for performance and promote from within were very useful concepts which have come handy.

"I wanted to do something for my country. There is more of that than anything else. Opportunities come and go, but your personal committment carries you through."



The Spread of Tandon

International Sales/Service/ Marketing Engineering Support Application Engineering International Purchase Office	Manufacturing-Exports Product Development	Indian Market Sales/ Service Marketing Manufacturing
Tandon Associates, C. A., USA	Ultra Tek devices LTD., Bombay-SEEPZ	Memory Electronics, Nasik
Tandon Associates, UK	Tancom Electronics, Bombay-SEEPZ	Tandon Liason Office, New Delhi
Golden Systems INC., C. A., USA	Advance Technology Devices, Bombay-SEEPZ	Pentax Engineering, Bombay
Tandon Associates PTE. LTD., Singapore	Tandon Motors PVT. LTD., Bombay-SEEPZ	Golden Computers, Bombay
Tantek Magnetics C. A., USA	Modular Electronics (I) PVT. LTD., Madras-MEPZ	Ideal Electronics, Indore
	Ultra Tek Devices LTD., Madras-MEPZ	
	Golden Systems Inc., Sri Lanka	
	Tancom Electronics, NEPZ	

Of the Rs 215 crore turnover this year, the highest grosser within the group was Ultra Tek Devices Ltd. (SEEPZ) which posted sales of Rs 75.52 crore (Rs 75.39 crore: exports, Rs 0.11 crore: deemed exports and Rs 0.012 crore: domestic sales; up from Rs 58.00 crore in 1992-93). Advance Technology Devices, with Rs 68.93 crore sales (all exports; up from Rs 40.09 crore in 1992-93) and Tancom Electronics with Rs 47.16 crore (up from Rs 23.35 crore in 1992-93) were next in succession. Other units registered varying results, but, most of them had a positive growth curve.

SEEPZ as the main base

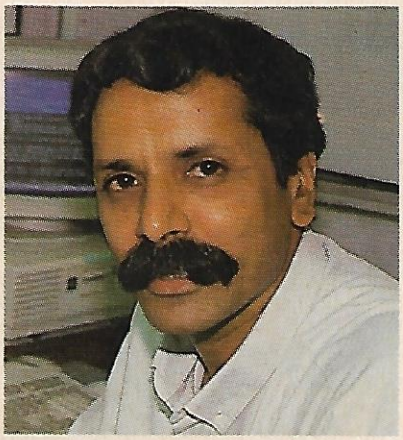
When the Tandons commenced operations, computer business was the domain of very large companies with sound R&D."We realised the potential of cost-effective infrastructure in India and selected SEEPZ as the site to make low-cost products. I had a personal commitment to prove to the world that Indians could also make internationally-competitive products," reminisces Tandon.

Double-sided heads, patented by Tandon, US, also were produced here and for the first few years brought in the major portion of grossings. A follow-on from these, the most important component of the floppy disk drive, was the stepper motor. Technology for this was obtained from Japan Servo Co. Ltd., a Hitachi subsidiary, and Tandon Motors Pvt. Ltd. (the facility for which in its present state can be stretched to produce 1.5 million motors per year) was formed for making this next most important part of the FDD. Falling demands for stepper motors, which were going out of style worldwide by replacement with voice coils, made this business go down. This made the Tandons expand their user base by making the motors not just for computers but for gas chromtography and other engineering devices. K.P. Murali, general manager, Tandon Motors Pvt. Ltd. (which produced 1,12,476 units in 1992) insists "the stepper motor technology will be phased out of use after the next 10 to 15 years."

Then came brushless DC motors, 20,000 units per month which this faciltiy makes, (rather insignificant compared to the 10 million miniature pieces churned out per day by Mubuchi of Japan) are supplied for tape drives, XY plotters and medical equipment.

"Life cycles of products are highly fluctuating and the company has to be prepared to face it."

S.C. Mathur
Asst. General Manager, Tandon.



Empowering the PC

Today, ninety percent of Tandon's power supplies go to the PC market. Switched mode power supplies of 10 to 300 watt capacity are made at the SEEPZ (monthly turnover of 200 K units) and at the MEPZ unit (75,000 per month). Tandon has shipped custom-designed units to IBM, Olivetti, Compaq, Nokia Data/International Computers Ltd., HP and Fujitsu and so on. Ultra-Tek Devices Ltd., the company which churned out more than 10,75,000 units in 1993-94 (compared with 10,023 in 1985-86 and 1,47,143 in 1990-91), has approvals from safety agencies like UL Compass, TUV-PSQA, ISO-9002, NORDIC, etc.

Tandons obtain about 30 percent of the components from within the country, including sheet metals parts and heat sinks. However, Kulkarni does not find "the basic componets like capacitor and resistor industry developing here for sometime."

A 40 watt power supply, for use in satellite receivers and works, is the most recent addition to this range. Full-scale production of this is slated to begin in August this year. "A direction towards UPS production" is taking shape at Tandon Motors and its designing is slated to begin by the fourth quarter of this year.

Onto the 1 GB head stack

Manufacture of the Winchester head stack commenced in the early 1980s. A 2 MB drive model—for 5 1/4 inch HDD, the TM

Tandon's Portfolio

Tancom Electronics
Started April 1982, SEEPZ, Bombay
Area: 40,000 sq. ft.

Exports in 1993-94:	Rs. **48.35** crore
Domestic Sales in 1993-94:	Rs. **00.39** crore
Sales in 1993-94:	Rs. **48.74** crore
Sales in 1992-93:	Rs. **23.00** crore

Products: Winchester Head Stacks (non-contact recording geads), critical read-write component in the hard disk drive and voice coils.

Growth in Sales: **111.9%**

Capacity per annum: 450,000 Head Stacks
7,200,000 Voice Coils

Customer Base: Seagate, Western Digital, Conner in the US and Singapore, Syquest Inc.

Advance Technology Devices
Started December 1985, SEEPZ, Bombay
Area: 60,000 sq. ft.

Sales in 1993-94:	Rs. **70.00** crore
Sales in 1992-93:	Rs. **40.09** crore

Products: Winchester Head Stacks

Capacity per annum: 750,000

Growth in Sales: **74.8%**

Customer Base: IBM, Seagate, Western Digital, Syquest

Tandon Motors Pvt. Ltd.
Started July 1981, SEEPZ, Bombay
Area: 21, 000 sq. ft.

Exports in 1993-94:	Rs. **0.47** crore
Domestic Sales in 1993-94:	Rs. **0.17** crore
Sales in 1993-94:	Rs. **0.64** crore
Sales in 1992-93:	Rs. **0.26** crore

Products: Stepper Motors (in technical collaboration with Japan Servo Co. Ltd., a subsidiary of Hitachi)

Growth in Sales: **146.2%**

Capacity per annum: 12,000,000 units

Customer Base: Kollmorgen, Tandon USA and Modi Xerox

Exports to: USA, Europe, Japan, Singapore

Golden Computers Pvt. Ltd.

Sales in 1993-94:	Rs. **6.58** crore
Sales in 1992-93:	Rs. **5.11** crore

Products: Computer Systems

Growth in Sales: **28.8%**

Ultra Tek Devices Ltd.
Started Aug. 1985, SEEPZ, Bombay and EPZ, Madras
Area: 100, 000 sq. ft., Bombay; 35,000 sq. ft., Madras

Exports in 1993-94:	Rs. **76.17** crore
Domestic Sales in 1993-94:	Rs. **00.13** crore
Sales in 1993-94:	Rs. **76.30** crore
Sales in 1992-93:	Rs. **58.00** crore

Products: Switching Mode Power Supplies, DC to DC Convertors, AC Adaptors

Growth in Sales: **31.55%**

Capacity per annum: 3,000,000 units of SMPS (Bombay), 1,000,000 units of SMPS (Madras)

Customer Base: IBM, Compaq, Olivetti, Golden Power, Fujitsu, GI Instruments & OEMs

Modular Electronics (I) Pvt. Ltd.
Started March 1986, EPZ, Madras
Area: 35,000 sq. ft.

Sales in 1993-94:	Rs. **8.52** crore
Sales in 1992-93:	Rs. **7.68** crore

Products: Winchester Head Stacks, Voice Coils and Power Supplies

Growth in Sales: **10.9%**

Capacity per annum: 350,000 Head Stacks, 5,4000,000 Voice Coils, 1,080,000 Power Supplies

Customer Base: IBM, Compaq, Seagate, Western Digital, Syquest, Maxtor

Eastern Peripherals Ltd.
Started 1982, SFFPZ, Bombay
Area: 35,000 sq. ft.

Sales in 1993-94:	Rs. **9.90** crore
Sales in 1992-93:	Rs. **24.53** crore

Products: Floppy Disk Drives, Computer Systems

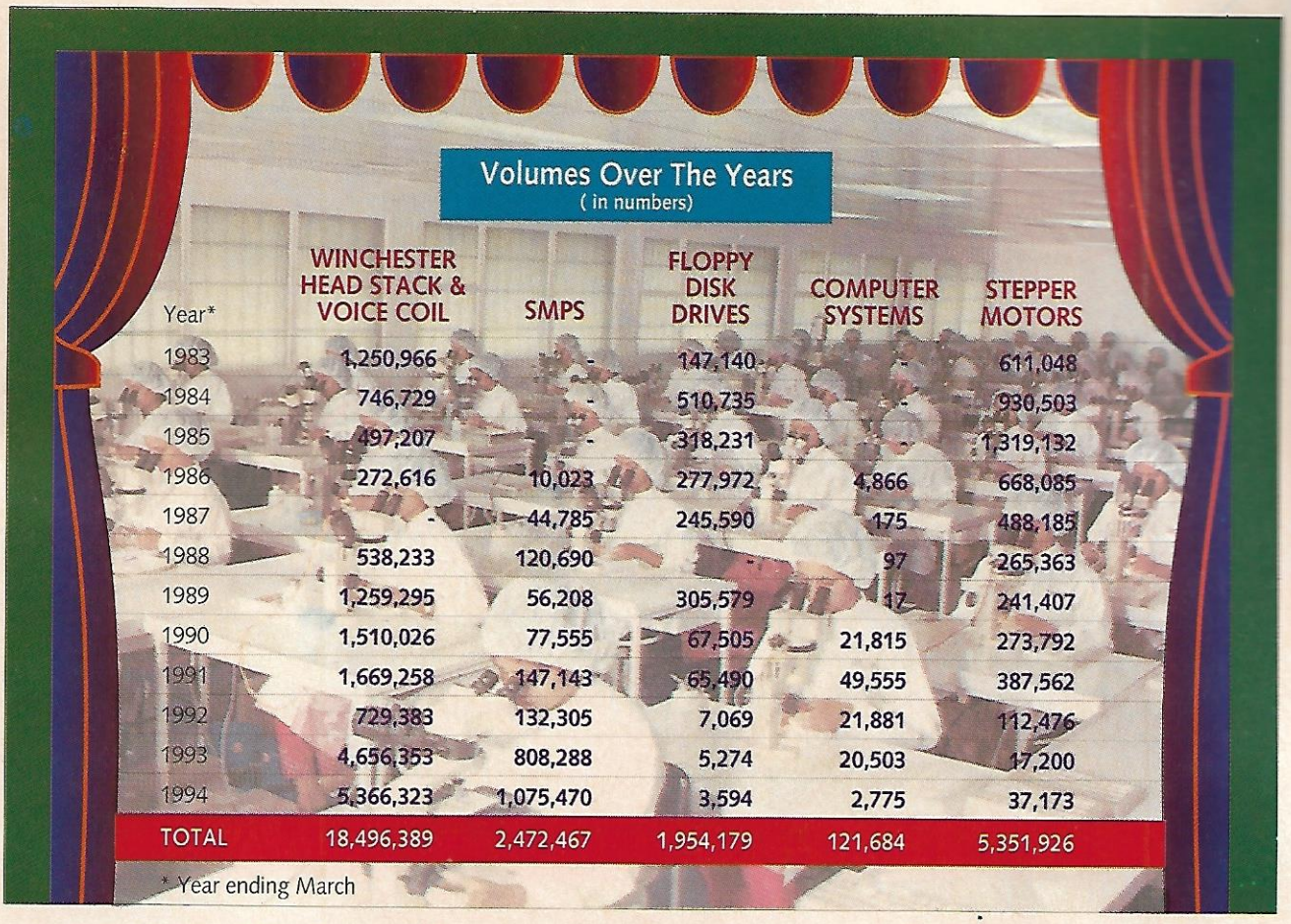
Growth in Sales: **-59.6%**

Capacity per annum: 500,000 FDD
100,000 systems

Customer Base: Tandon, USA; UK and OEMs

Chetu B.K. Sharma

Charts by B. K. Sharma



Volumes Over The Years
(in numbers)

Year*	WINCHESTER HEAD STACK & VOICE COIL	SMPS	FLOPPY DISK DRIVES	COMPUTER SYSTEMS	STEPPER MOTORS
1983	1,250,966	-	147,140	-	611,048
1984	746,729	-	510,735	-	930,503
1985	497,207	-	318,231	-	1,319,132
1986	272,616	10,023	277,972	4,866	668,085
1987	-	44,785	245,590	175	488,185
1988	538,233	120,690	-	97	265,363
1989	1,259,295	56,208	305,579	17	241,407
1990	1,510,026	77,555	67,505	21,815	273,792
1991	1,669,258	147,143	65,490	49,555	387,562
1992	729,383	132,305	7,069	21,881	112,476
1993	4,656,353	808,288	5,274	20,503	17,200
1994	5,366,323	1,075,470	3,594	2,775	37,173
TOTAL	18,496,389	2,472,467	1,954,179	121,684	5,351,926

* Year ending March

262, was the first to roll out. Next came the 40 MB, followed by the 80 MB and in 1989 came the 120 MB model. The subsequent year saw the launch of the 340 MB model (production of which will cease anytime now), with 540 MB on its tail and the 1 GB, to be in production this month, is the latest in line. "The 80 MB model was the longest—two years," says Mathur. Floppy Disk Drives production, with voice coils, were the Tandons' next venture. The associate base in the US "pioneered the revolution of lowest cost FDDs," revealed Tandon.

Tandon's fortunes took a dip in 1991-92. Contributing to this was the forex dilemma the country was facing. "We could not get any foreign exchange to buy raw material," recalls Tandon. Without the forex component, import-intensive exports of companies, like this, suffered. This impaired the Tandon's reputation with customers who "are not concerned about any domestic policy or crisis but want their demands met as scheduled," rues a GM.

Customers evaluate services on the basis of plus-minus one day on delivery schedules.

Lone Ranger

But, today, the Tandons have stories of successes aplenty. And, what about the future? In 1994-95 the heads business is gauged to grow by 15 to 20 percent and power supplies by 30 percent. Motors is one product which could take a quantum leap of 200 to 300 percent. Total revenues, however, will hinge mostly and equally on SMPS and heads, with a lesser percentage coming from motors. Tandon does not rule out a foray into the monitors and keyboards business even though "currently there is no space."

The Tandons have shown a clear-cut methodology to success in this field. Their example, as shown, is not easy to emulate.

However, what is reconfirmed is that India presents ample opportunities as a viable manufacturing base for labour-intensive production, provided that quality is not compromised on, and one eye is constantly trained on the market drift. The Asian Tigers are what they are today because they worked on these very base—with cost-effective infrastructure and labour as the rock bottomline, they have built themselves into world-class manufacturers. If the Tandon example is taken as a role model and a whole industry mushrooms on these very lines, it would be a big step towards making India an elephant among the Asian Tigers.